Evidential Dirichlet Fusion for Trusted Multimodal Spectrum Sensing

Jun Liu
Minzu University of China
Beijing, China
24011044@muc.edu.cn

***Abstract*—Reliable spectrum sensing under low signal-to-noise ratio (SNR) and unseen modulation conditions remains challenging because existing deep learning detectors often rely on a single representation and produce overconfident decisions under distribution shifts. This paper proposes an uncertainty-aware multimodal framework for robust binary spectrum sensing. It fuses complementary temporal and signal-image representations through gated cross-attention and employs Dirichlet evidential learning to jointly estimate spectrum-occupancy probabilities and epistemic uncertainty. Experiments on the RadioML 2018.01A dataset cover in-distribution validation, unseen-SNR testing, and unseen-modulation testing. Experimental results demonstrate that the proposed method consistently outperforms representative single-modal and multimodal baselines across all three evaluation protocols, improving sensing accuracy by approximately 1–2 percentage points while maintaining a lower false-alarm rate and robust performance under unseen-SNR and unseen-modulation conditions. These results demonstrate that uncertainty-aware multimodal evidence fusion provides a more reliable and trustworthy solution for practical spectrum sensing under distribution shifts.**

Keywords—Spectrum sensing, cognitive radio, multimodal learning, evidential deep learning, uncertainty estimation.

I. INTRODUCTION

Spectrum scarcity remains a persistent challenge in modern wireless communication systems. The rapid development of cognitive radio, dynamic spectrum access, the Internet of Things, and next-generation wireless networks has substantially increased the demand for limited radio resources, making conventional static spectrum allocation increasingly inefficient. More flexible spectrum-management mechanisms are thus required to identify temporarily available frequency bands and enable secondary users to access them without causing harmful interference to licensed users. As a fundamental component of this process, spectrum sensing determines whether a target band is idle or occupied and directly affects the efficiency and reliability of dynamic spectrum access [1]–[3].

Binary spectrum sensing is commonly formulated as a hypothesis-testing problem. Under H_0, the licensed user is absent and the received observation contains only noise, whereas under H_1, the licensed user is present and the observation contains both signal and noise. A reliable detector should identify weak primary-user signals while maintaining a low false-alarm probability. In practical wireless environments, this requirement is complicated by propagation loss, fading, interference, hardware impairments, noise uncertainty, and modulation diversity. Moreover, the SNR ranges and modulation formats encountered during deployment may differ substantially from those observed during training. Practical spectrum sensing should consequently be regarded not only as an in-distribution classification task, but also as a robustness and generalization problem under signal-distribution shifts [3], [11].

Conventional sensing techniques, including energy detection, matched filtering, and cyclostationary feature detection, laid the foundation for early cognitive radio systems. Despite their important role, these methods generally depend on fixed statistical assumptions or prior knowledge of the transmitted signal. Energy detection is vulnerable to noise uncertainty and threshold mismatch, matched filtering requires accurate waveform information and synchronization, and cyclostationary detection often incurs considerable computational cost and relatively long observation intervals. Their performance may therefore deteriorate when practical channel conditions deviate from the underlying assumptions, particularly in low-SNR environments [3], [4].

Against this background, learning-based spectrum sensing has attracted increasing attention as a means of reducing dependence on handcrafted decision rules and signal-specific prior knowledge. Deep learning methods can learn discriminative representations directly from IQ samples, spectral correlation functions, or signal-derived images. CNN-based detectors are effective in extracting spectral and spatial patterns, whereas recurrent architectures are suitable for modeling sequential dependencies and temporal evolution [5]–[7]. Nevertheless, most existing approaches rely primarily on a single signal representation. Temporal features may be severely distorted by strong noise, while spectral or image-based representations may not fully preserve fine-grained temporal dynamics. These complementary strengths and limitations motivate the joint modeling of heterogeneous signal representations, particularly under weak-signal and distribution-shift conditions.

A recent multimodal spectrum-sensing approach, BCAM-Net, addresses single-view limitations by jointly modeling CWT-based time-frequency representations and magnitude-phase temporal features. Its bidirectional cross-attention module allows the temporal and time-frequency streams to alternately serve as queries and keys/values, enabling mutual calibration and adaptive token-level reweighting. The resulting fused representation combines the original global time-frequency descriptor, its cross-attention-enhanced counterpart, and a time-frequency-guided temporal summary before binary classification [8]. Although this design improves bidirectional feature interaction and robustness under generalized Gaussian noise, the attention-enhanced features are incorporated into the fused representation without an additional gating stage dedicated to regulating the magnitude of cross-modal updates. Moreover, its MLP–Softmax classifier does not explicitly represent the evidence supporting a prediction or quantify epistemic uncertainty. An important remaining issue is therefore how to retain the advantages of bidirectional cross-modal calibration while providing finer control over feature updates and more trustworthy decisions under distribution shifts.

Decision reliability is particularly important because representative deep spectrum-sensing models commonly produce deterministic probability outputs [5]–[8]. Modern neural networks may generate poorly calibrated confidence estimates even when their classification accuracy is high [10]. This problem can become more pronounced when the testing distribution differs from the training distribution, since uncertainty estimation and probability calibration may degrade under dataset shift [11]. In spectrum sensing, such behavior may result in unwarranted confidence for weak, ambiguous, or previously unseen signals.

Post-hoc calibration methods, such as temperature scaling, can improve confidence calibration using a held-out validation set [10], but they do not explicitly model the evidence associated with individual inputs and may become less reliable when the deployment distribution changes [11]. Bayesian approximations can estimate epistemic uncertainty, yet practical implementations such as Monte Carlo dropout generally require repeated stochastic forward passes during inference [12]. In contrast, evidential learning predicts nonnegative class evidence and uses it to parameterize a Dirichlet distribution over class probabilities. The total Dirichlet concentration reflects the amount of evidence supporting a prediction, allowing low-evidence inputs to be associated with greater uncertainty rather than unwarranted confidence [9]. This direct relationship between evidence strength and decision uncertainty makes evidential modeling well suited to spectrum sensing under SNR and modulation distribution shifts.

Motivated by these observations, the proposed framework addresses three corresponding design challenges.

At the representation level, the incomplete characterization provided by a single signal view is addressed by constructing complementary temporal and signal-derived image representations from the same IQ observation. Dedicated branches model waveform evolution and phase-frequency dynamics, as well as spectral, time-frequency, state-transition, and recurrence structures.

At the fusion level, bidirectional cross-attention facilitates mutual calibration between temporal and image-based representations, but the resulting feature updates may contribute differently across inputs. To provide finer control over this process, the proposed framework augments bidirectional cross-attention with modality-specific learnable gates. The attention operations capture complementary dependencies, whereas the gates adaptively control the amount of cross-attended information incorporated into the temporal and image representations. This design is intended to preserve useful cross-modal interactions while limiting the influence of potentially misleading updates.

At the decision level, deterministic Softmax outputs cannot explicitly indicate whether a prediction is supported by sufficient evidence. To address this limitation, the fused representation is mapped to Dirichlet evidence, from which spectrum-occupancy probabilities and an evidence-based estimate of epistemic uncertainty are jointly derived [9]. Weak, ambiguous, or previously unseen signals can therefore be represented by lower evidence and higher uncertainty rather than being assigned unjustifiably confident predictions.

The main contributions of this work are summarized as follows:

- **Dual-branch multimodal representation**: Complementary temporal and signal-derived image representations are jointly modeled to provide a more comprehensive characterization of weak and previously unseen signals than single-view detectors.
- **Gated bidirectional cross-modal fusion**: A gated bidirectional cross-attention mechanism is developed to retain two-way complementary interactions while adaptively controlling the magnitude of cross-modal feature updates.
- **Uncertainty-aware decision modeling**: Dirichlet evidential learning is incorporated to jointly derive spectrum-occupancy probabilities and an evidence-based estimate of epistemic uncertainty, thereby mitigating unwarranted confidence under unfamiliar signal conditions.
- **Evaluation under distribution shifts**: Extensive experiments on the RadioML 2018.01A dataset are conducted under in-distribution validation and generalization to unseen SNRs and modulation formats. The results demonstrate overall performance improvements over representative single-modal and multimodal baselines, particularly under challenging distribution shifts.

II. Related Work

A. Conventional Spectrum Sensing

Conventional spectrum sensing methods infer spectrum occupancy through statistical characteristics of the received signal. Energy detection estimates the received energy over an observation interval and compares it with a predefined threshold. Because it requires neither waveform knowledge nor synchronization, energy detection offers low implementation complexity and is widely applicable to blind sensing. Its effectiveness, however, depends directly on the accuracy of the noise-power estimate. At low SNRs, the energy distributions under H_0 and H_1 become difficult to separate, causing threshold mismatch and noise uncertainty to produce substantial detection errors [3].

Matched filtering exploits a different principle by correlating the received samples with a known primary-user waveform. Accurate waveform and synchronization information enables coherent accumulation of signal energy and therefore yields high detection sensitivity with relatively short observations. This advantage is obtained at the cost of strong prior assumptions: the signal format, pulse shape, timing, and often channel characteristics must be known in advance. Its applicability is consequently limited in heterogeneous or blind sensing scenarios in which primary-user parameters are unavailable or vary over time [3].

Cyclostationary feature detection uses periodic statistical structures introduced by modulation, symbol timing, pilots, or coding. Unlike energy detection, it can distinguish modulated signals from stationary noise by examining cyclic correlations or spectral correlation features, which improves weak-signal discrimination under some low-SNR conditions. Lundén et al. exploited multiple cyclic frequencies to strengthen the sensing statistic and improve robustness to individual feature variations [4]. Nevertheless, cyclostationary sensing generally requires long observation windows, accurate cyclic-feature selection, and computationally intensive correlation estimation. Thus, conventional methods achieve reliable performance only when their assumptions regarding noise statistics, signal templates, or cyclic structures remain valid, motivating the transition toward data-driven sensing.

B. Learning-Based Spectrum Sensing

Learning-based methods replace fixed analytical statistics with classifiers learned from observed data. Early machine-learning schemes typically used energy, covariance, entropy, or spectral descriptors as inputs to support vector machines and shallow neural networks. These approaches provide greater flexibility than fixed-threshold detectors, but the attainable performance remains bounded by the discriminative quality of manually designed features.

Deep learning reduces this dependence by learning task-oriented representations from sensing observations. Lee et al. proposed a CNN-based cooperative sensing framework that learns how to combine observations collected by multiple secondary users [5]. Convolutional processing enables the detector to exploit correlations among distributed measurements rather than relying on a manually derived fusion rule. The method improves cooperative detection under the conditions represented during training, but it remains tied to a predefined multiuser observation structure and produces a deterministic closed-set decision.

Tekbiyik et al. combined the spectral correlation function with a CNN for joint spectrum sensing and signal identification [6]. The spectral correlation function embeds physically meaningful cyclostationary patterns into a two-dimensional representation, while the CNN learns discriminative local structures that would be difficult to specify through fixed cyclic-frequency rules. This combination improves resistance to ordinary noise and supports signal identification; nevertheless, the learned representation remains dependent on a single predefined transform, and its performance may degrade when the spectral-correlation patterns differ from those represented in the training set.

Sequence models offer an alternative by operating directly on temporal observations. Kumar et al. employed an RNN–BiLSTM detector for NOMA waveforms under diverse channel conditions [7]. Bidirectional recurrence captures dependencies from both past and future samples, allowing the model to exploit waveform evolution that is unavailable to memoryless classifiers. The evaluation nevertheless remains based on predefined waveform and channel configurations, while the temporal branch alone may become unreliable when weak signal dynamics are heavily obscured by noise.

Image-based approaches instead transform IQ observations into structured spectral representations. The STFT–CNN method converts the received sequence into a time-frequency map and uses convolutional filters to detect localized spectral patterns [13]. A wavelet–ResNet detector further uses multiscale CWT representations to retain transient and frequency-dependent structures, improving low-SNR discrimination [18]. These transforms make weak spectral patterns more accessible to convolutional networks, but a single image representation may compress fine temporal, phase, and instantaneous-frequency information. Attention-based models such as Spectrum Transformer extend learned sensing to long-range wideband dependencies and improve inter-band feature modeling [17]. Despite these advances, most learning-based methods remain single-view classifiers evaluated under predefined signal and noise distributions, leaving their behavior under unseen SNRs, unseen modulation formats, and other distribution shifts insufficiently characterized.

C. Multimodal Spectrum Sensing

Multimodal spectrum sensing has emerged to overcome the information loss associated with a single signal view. Instead of selecting either raw temporal samples or transformed spectral images, these methods construct parallel branches and combine representations that describe different physical properties of the same observation.

Gao et al. proposed a GAN–GRU–YOLO framework for consumer-IoT spectrum sensing [16]. A CWT-based spectrogram branch uses YOLO to extract high-order time-frequency patterns, while GRU and CNN components model raw sequential characteristics; a GAN is further introduced to augment training data. The complementary branches improve feature coverage and achieve favorable detection performance in the evaluated IoT scenarios. The resulting architecture is relatively complex, however, and its performance deteriorates markedly in the lowest-SNR region, indicating that multimodal feature availability alone does not ensure reliable fusion when both representations are strongly corrupted.

Fang et al. combined CNN and Transformer components for cooperative spectrum sensing [15]. The CNN captures local structures from individual secondary-user observations, whereas the Transformer models temporal and inter-user dependencies over longer ranges. This hybrid design improves cooperative detection by jointly exploiting local and global correlations, but its computational cost is higher than that of compact convolutional detectors. Moreover, the method focuses on correlations among cooperative observations rather than explicitly controlling the interaction between distinct physical signal modalities.

A more direct form of multimodal fusion was introduced in TFCFN, which combines a GRU-based temporal branch with an FFT–CNN frequency-domain branch [18]. Cross-attention allows one representation to query complementary information from the other, thereby improving sensing under generalized Gaussian noise. The fusion is primarily unidirectional, however, so one stream is conditioned on the other without symmetric refinement. When impulsive noise distorts either modality, this asymmetric interaction may propagate modality-specific artifacts and limit fusion stability.

BCAM-Net extends this line of research through bidirectional cross-attention between CWT-based time-frequency tokens and magnitude–phase temporal features [8]. By alternately assigning the two streams as queries and keys/values, the model enables mutual calibration and adaptive token-level reweighting. The original global time-frequency descriptor, the cross-attention-enhanced time-frequency descriptor, and the time-frequency-guided temporal summary are then concatenated for classification. This bidirectional design improves low-SNR sensing under generalized Gaussian noise and addresses the asymmetry of one-way fusion. Nevertheless, the attention-enhanced features enter the fused representation without an additional modality-specific gating stage dedicated to controlling the magnitude of cross-modal updates. More importantly, the final MLP–Softmax classifier still produces deterministic probabilities without explicitly representing the amount of evidence supporting each decision.

The development from parallel branches to cross-attention and bidirectional calibration demonstrates the value of complementary signal views. Yet existing multimodal detectors are primarily optimized for feature discrimination under predefined training distributions. Their deterministic outputs may remain overconfident when SNR ranges, modulation formats, or signal statistics differ from those observed during training [1], [3]. Evidential deep learning offers a potential remedy by parameterizing a Dirichlet distribution from nonnegative class evidence, thereby associating weak evidence with increased predictive uncertainty [9]. Its integration with gated multimodal spectrum sensing, however, remains largely unexplored. This gap motivates the proposed framework, which combines complementary temporal and signal-derived image representations, gated bidirectional cross-modal updates, and evidence-based decision modeling within a unified sensing architecture.

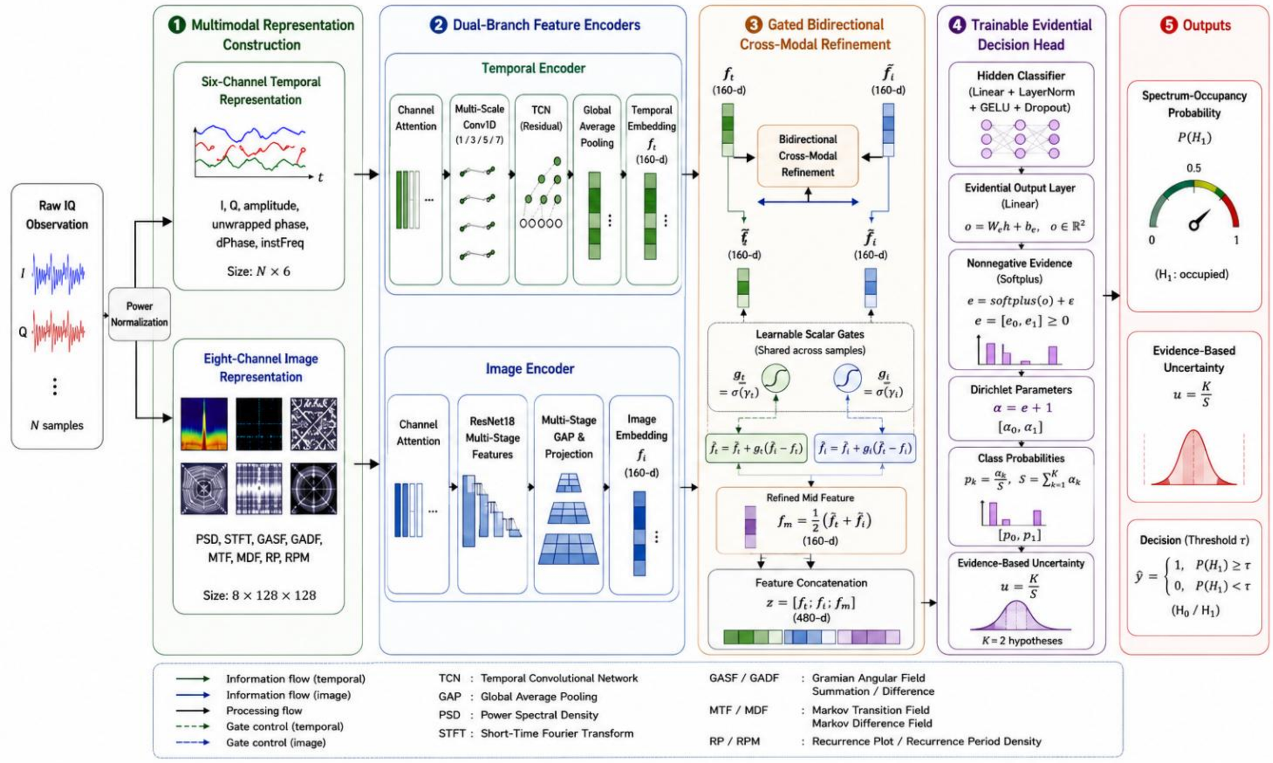


Fig. 1. Overall architecture of the proposed uncertainty-aware multimodal spectrum-sensing framework. A raw IQ observation is power-normalized and converted into complementary temporal and signal-derived image representations. The two representations are processed by dedicated feature encoders, refined through gated bidirectional cross-modal interaction, and mapped to Dirichlet evidence for spectrum-occupancy prediction and uncertainty estimation.

III. Framework Overview

A. Algorithm Overview

The proposed framework is developed for reliable binary spectrum sensing under low signal-to-noise ratio (SNR) conditions and signal-distribution shifts. Given a complex baseband IQ observation $\mathbf{x} = [\mathbf{I}, \mathbf{Q}] \in \mathbb{R}^{N\times 2}$, where $N = 1024$, the sensing task is formulated as

$$\mathbf{x}[n] = \begin{cases} \mathbf{w}[n], & H_0, \\ \mathbf{s}[n] + \mathbf{w}[n], & H_1, \end{cases} n = 1, \ldots, N, \tag{1}$$

where $\mathbf{s}[n]$ denotes the received primary-user signal and $\mathbf{w}[n]$denotes additive noise. Under H_0, the observed spectrum segment is regarded as idle, whereas under H_1, a primary-user signal is present. The objective is to detect weak occupied-spectrum observations while maintaining a low false-alarm probability.

Before multimodal representation construction, each IQ observation is normalized by its average power:

$$\bar{\mathbf{x}} = \frac{\mathbf{x}}{\sqrt{\frac{1}{2N}\sum_{n=1}^{N}(I^2[n] + Q^2[n]) + \varepsilon}}, \tag{2}$$

where $\varepsilon$is a small positive constant introduced for numerical stability.

As illustrated in Fig. 1, the proposed framework contains four main stages. First, the normalized IQ sequence is converted into a six-channel temporal representation and an eight-channel signal-derived image representation. Second, a temporal convolutional encoder and a residual image encoder extract complementary feature embeddings from the two modalities. Third, a gated bidirectional cross-modal refinement module transfers information between the two embeddings while regulating the strength of the residual updates. Finally, the original and refined representations are processed by a trainable evidential decision head, which produces nonnegative class evidence, spectrum-occupancy probabilities, and an evidence-based uncertainty estimate.

like conventional dual-branch architectures that directly concatenate independently extracted features, the proposed framework introduces controlled bidirectional feature refinement before decision making. Moreover, the trainable evidential head distinguishes relative class preference from the amount of evidence supporting the prediction. All trainable components, including the two feature encoders, the cross-modal refinement module, the hidden classifier, and the evidential output layer, are jointly optimized through the end-to-end objective introduced in Section III-F.

B. Multimodal Signal Representation

(1) Temporal Representation

The temporal representation organizes each normalized IQ observation into six physically meaningful channels. Let $I[n]$ and $Q[n]$ denote the in-phase and quadrature components, respectively. The amplitude, unwrapped phase, phase difference, and normalized instantaneous frequency are calculated as

$$\begin{aligned} A[n] &= \sqrt{I^2[n] + Q^2[n]}, \\ \phi[n] &= \mathrm{unwrap}(\mathrm{atan2}(Q[n], I[n])), \\ \Delta\phi[n] &= \phi[n] - \phi[n-1], \\ f_{\mathrm{inst}}[n] &= \mathrm{clip}\left(\frac{\Delta\phi[n]}{2\pi}, -1, 1\right). \end{aligned} \tag{3}$$

The temporal input is then expressed as

$$\mathbf{X}_t = [\mathbf{I}, \mathbf{Q}, \mathbf{A}, \boldsymbol{\phi}, \Delta\boldsymbol{\phi}, \mathbf{f}_{\mathrm{inst}}] \in \mathbb{R}^{N\times 6}. \tag{4}$$

Each channel is standardized independently using the mean and standard deviation of the current observation. This compact representation preserves the original sampling order while explicitly exposing amplitude and phase-frequency evolution to the temporal encoder. Because this representation mainly serves as a structured organization of the IQ sequence rather than a methodological contribution, no further derivation is introduced.

(2) Signal-Derived Image Representation

The same IQ observation is further transformed into an eight-channel image tensor:

$$\mathbf{X}_i = \mathrm{Stack}(\mathbf{P}_{\mathrm{PSD}}, \mathbf{P}_{\mathrm{STFT}}, \mathbf{G}_{\mathrm{S}}, \mathbf{G}_{\mathrm{D}}, \mathbf{M}_{\mathrm{T}}, \mathbf{M}_{\mathrm{D}}, \mathbf{R}_{\mathrm{P}}, \mathbf{R}_{\mathrm{PM}}) \in \mathbb{R}^{8\times H\times W}, \tag{5}$$

where $H = W = 128$ in the implementation.

The power spectral density (PSD) and short-time Fourier transform (STFT) channels characterize global spectral-energy distribution and local time-frequency variation, respectively. Specifically, Welch estimation is first applied to obtain the logarithmic PSD vector $\mathbf{p}$. After min-max normalization, a two-dimensional PSD relation map is constructed as

$$\mathbf{P}_{\mathrm{PSD}} = \tilde{\mathbf{p}}\,\tilde{\mathbf{p}}^{\mathrm{T}}, \tag{6}$$

where $\tilde{\mathbf{p}}$ denotes the normalized logarithmic PSD vector. The outer-product construction converts the one-dimensional spectrum into a second-order relation map, thereby exposing interactions between different frequency positions. The STFT magnitude map is used directly after logarithmic scaling, normalization, and spatial resizing. Because PSD and STFT are established spectral representations, they are not discussed further.

To complement conventional spectral images, Gramian angular fields encode the temporal ordering and pairwise relationships of the amplitude sequence. The downsampled amplitude sequence is first normalized to $[-1,1]$ and mapped into angular coordinates:

$$\theta_n = \arccos(\tilde{A}[n]). \tag{7}$$

The Gramian angular summation field (GASF) and Gramian angular difference field (GADF) are then defined as

$$\begin{aligned} G_{\mathrm{S}}(m,n) &= \cos(\theta_m + \theta_n), \\ G_{\mathrm{D}}(m,n) &= \sin(\theta_m - \theta_n). \end{aligned} \tag{8}$$

Since the row and column indices correspond to temporal positions, the Gramian mappings retain the ordering of the original sequence while encoding relationships between distant samples [19]. Compared with a conventional spectrogram, which mainly represents localized energy distribution, GASF and GADF provide complementary descriptions of long-range amplitude evolution and relative temporal behavior.

The Markov transition field (MTF) describes transitions between discretized amplitude states. After quantizing the amplitude sequence into $B = 8$ states, its elements are defined as

$$M_{\mathrm{T}}(m,n) = P(q_n \mid q_m), \tag{9}$$

where q_m and q_n denote the quantized amplitude states at positions m and n, respectively. A row-normalized transition-density map, termed the marginal-distribution field (MDF) in this work, is further constructed as

$$M_{\mathrm{D}}(m,n) = \frac{M_{\mathrm{T}}(m,n)}{\sum_j M_{\mathrm{T}}(m,j) + \varepsilon}. \tag{10}$$

The MTF and MDF channels emphasize relative transition behavior rather than absolute spectral magnitude.

Finally, the recurrence channels encode repeated amplitude states at two resolutions:

$$\begin{aligned} R_\rho(m,n) &= \mathbb{I}(\mid A[m] - A[n] \mid \le \epsilon_\rho), \\ \rho &\in \{5\%, 20\%\}, \end{aligned} \tag{11}$$

where ϵ_ρ is selected to obtain the specified recurrence percentage. The 20% and 5% configurations form the RP and RPM channels, respectively, capturing relatively coarse and fine recurrence structures [20].

All eight image channels are independently normalized and resized to 128×128 before channel-wise stacking. By jointly encoding spectral energy, local time-frequency variation, pairwise temporal relationships, state transitions, and recurrence structures, the image modality provides a complementary view of the signal that cannot be obtained from the raw temporal sequence alone.

The GASF, GADF, and MTF representations were originally introduced for time-series imaging at IJCAI, where the mappings were designed to preserve sequential structure while enabling convolutional processing.

C. Dual-Branch Feature Extraction

(1) Temporal Feature Encoder

The temporal encoder first applies channel-wise recalibration to the six temporal channels. A parallel multi-scale one-dimensional convolutional front end with kernel sizes 1, 3, 5, and 7 then extracts waveform patterns over different local receptive fields. The concatenated 64-channel feature map is subsequently processed by two residual temporal convolutional network (TCN) blocks with dilation factors of 2 and 4.

Dilated temporal convolutions increase the effective receptive field without requiring recurrent computation, making them suitable for capturing both short-term waveform changes and longer phase-frequency dependencies [21]. Global average pooling over the temporal dimension produces the temporal embedding

$$\mathbf{f}_t = E_t(\mathbf{X}_t) \in \mathbb{R}^d, d = 160. \tag{12}$$

The internal structure of the TCN is kept concise because temporal convolution is an established sequence-modeling technique rather than the primary contribution of this work.

(2) Image Feature Encoder

The image encoder first performs channel recalibration over the eight signal-derived images. The reweighted tensor is then processed by an eight-channel ResNet18 backbone. Residual learning provides direct gradient paths and facilitates the optimization of hierarchical convolutional representations [22].

Instead of using only the deepest feature map, the proposed image encoder retains the outputs of the second, third, and fourth residual stages. After global average pooling, the resulting feature vectors have dimensions of 128, 256, and 512, respectively. Their concatenation forms a multi-scale pyramid representation:

$$\mathbf{f}_{\text{pyr}} = [\text{GAP}(\mathbf{F}_2); \text{GAP}(\mathbf{F}_3); \text{GAP}(\mathbf{F}_4)] \in \mathbb{R}^{896}. \tag{13}$$

A linear projection maps the pyramid representation into the shared embedding space:

$$\mathbf{f}_i = E_i(\mathbf{X}_i) = \text{Proj}(\mathbf{f}_{\text{pyr}}) \in \mathbb{R}^d. \tag{14}$$

The multi-stage aggregation preserves relatively fine signal-image structures together with higher-level global patterns. The use of ResNet18 itself is not treated as a methodological contribution.

The ability of TCNs to capture long-range dependencies through temporal convolutions and the optimization advantages of residual networks have been established in prior CVPR studies.

D. Gated Bidirectional Cross-Modal Refinement

The temporal and image encoders characterize complementary properties of the same IQ observation. Direct feature concatenation preserves both embeddings but does not allow information from one modality to refine the other before classification. Moreover, when one representation is strongly affected by noise or an unfamiliar modulation pattern, unconstrained cross-modal transfer may introduce irrelevant information into the fused representation.

Prior multimodal studies have shown that explicitly controlled information exchange can be more effective than simply combining final unimodal predictions or representations [23], [24]. Attention bottlenecks and multimodal token fusion, for example, restrict or selectively replace transmitted information to avoid indiscriminate multimodal mixing. In spectrum sensing, attention-based modeling has also been used to capture long-range spectral dependencies [17].

In the present implementation, both branches are globally pooled before fusion, and the cross-modal interaction is therefore performed at the embedding level. The image-to-temporal and temporal-to-image context vectors are calculated as

$$\begin{aligned} \mathbf{c}_{t\leftarrow i} &= \mathbf{W}_{vi}\mathbf{f}_i, \\ \mathbf{c}_{i\leftarrow t} &= \mathbf{W}_{vt}\mathbf{f}_t, \end{aligned} \tag{15}$$

where $\mathbf{W}_{vi}, \mathbf{W}_{vt} \in \mathbb{R}^{d\times d}$ are learnable projections. The first context vector transfers image-derived structural information to the temporal representation, whereas the second transfers temporal evolution information to the image representation.

Two learnable scalar gates regulate the strengths of the corresponding residual updates:

$$g_t = \sigma(\gamma_t), g_i = \sigma(\gamma_i), \tag{16}$$

where γ_t and γ_i are trainable parameters and $\sigma(\cdot)$ denotes the sigmoid function. Since these gates are shared across samples, they learn the overall contribution of cross-modal information rather than estimating sample-specific modality reliability.

The refined embeddings are obtained through gated residual updates:

$$\begin{aligned} \tilde{\mathbf{f}}_t &= \text{LN}(\mathbf{f}_t + g_t\mathbf{c}_{t\leftarrow i}), \\ \tilde{\mathbf{f}}_i &= \text{LN}(\mathbf{f}_i + g_i\mathbf{c}_{i\leftarrow t}), \end{aligned} \tag{17}$$

where $\text{LN}(\cdot)$ denotes layer normalization. The residual paths preserve the original modality-specific information, whereas the gates prevent the transferred context from being injected with an unconstrained magnitude.

The refined multimodal representation is defined as

$$\mathbf{f}_m = \frac{1}{2}(\tilde{\mathbf{f}}_t + \tilde{\mathbf{f}}_i) \in \mathbb{R}^d. \tag{18}$$

The final representation supplied to the evidential decision head is

$$\mathbf{z} = [\mathbf{f}_t; \mathbf{f}_i; \mathbf{f}_m] \in \mathbb{R}^{3d}. \tag{19}$$

This formulation preserves three types of information: the original temporal evidence, the original image evidence, and the mutually refined multimodal evidence. Retaining the original embeddings prevents the decision head from depending exclusively on transferred features, while the refined representation captures cross-modal consistency.

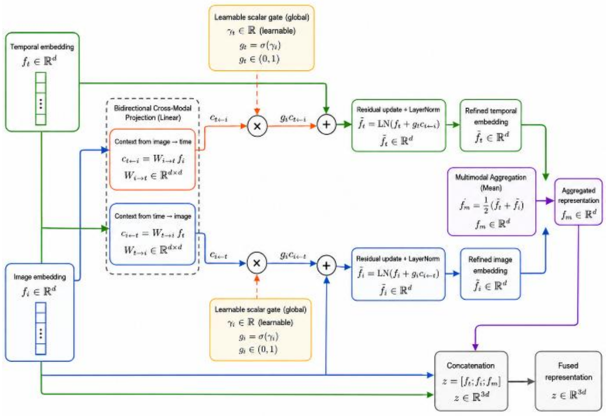


Fig. 2. Gated bidirectional cross-modal refinement module. The temporal and image embeddings are projected into two directional context vectors. Learnable scalar gates regulate the strengths of the residual updates, after which layer normalization and multimodal aggregation produce the refined representation.

E. End-to-End Dirichlet Evidential Decision Modeling

A conventional Softmax classifier produces normalized class probabilities regardless of whether the input provides sufficient support for a reliable decision. Consequently, weak, ambiguous, or distribution-shifted observations may still receive highly concentrated predictions. Evidential deep learning instead places a Dirichlet distribution over class probabilities and interprets the nonnegative neural outputs as evidence supporting the individual classes [9].

Given the combined representation $\mathbf{z}$, a trainable hidden classifier first produces

$$\mathbf{h} = \mathrm{MLP}_c(\mathbf{z}) \in \mathbb{R}^{256}. \tag{20}$$

The trainable evidential output layer then generates two evidence logits:

$$\mathbf{o} = \mathbf{W}_e\mathbf{h} + \mathbf{b}_e \in \mathbb{R}^K, K = 2, \tag{21}$$

where $\mathbf{W}_e$and $\mathbf{b}_e$are explicitly included in the optimizer and jointly updated with the upstream network.

To ensure nonnegative class evidence, the logits are transformed using the Softplus function:

$$e_k = \mathrm{softplus}(o_k) + \varepsilon, k \in \{0,1\}. \tag{22}$$

The evidence values parameterize a Dirichlet distribution through

$$\alpha_k = e_k + 1, S = \sum_{k=1}^{K} \alpha_k, \tag{23}$$

where $\alpha_k$denotes the concentration associated with class k, and Sdenotes the total Dirichlet strength.

The expected class probability, class belief mass, and remaining uncertainty mass are calculated as

$$p_k = \frac{\alpha_k}{S}, b_k = \frac{e_k}{S}, u = \frac{K}{S}. \tag{24}$$

They satisfy the subjective-opinion constraint

$$\sum_{k=1}^{K} b_k + u = 1. \tag{25}$$

The quantities in (23)–(25) have direct interpretations. The evidence $e_k$measures the support collected for class k, while Smeasures the total amount of evidence supplied by the current observation. When both class evidence values are small, Sremains low and the uncertainty mass $u$remains high. When consistent discriminative features produce strong evidence, Sincreases and $u$decreases.

This interpretation differs from Softmax normalization. Softmax only represents the relative preference between classes, whereas the Dirichlet formulation additionally indicates whether the preference is supported by sufficient evidence. Recent reliable multi-view studies have similarly used evidence distributions and uncertainty-aware integration to improve decision reliability when different views are noisy, inconsistent, or conflicting [25]–[26].

For binary spectrum sensing, the spectrum-occupancy probability is

$$p(H_1 \mid \mathbf{x}) = p_1 = \frac{\alpha_1}{\alpha_0 + \alpha_1}. \tag{26}$$

The scalar $u$is used as an evidence-deficiency indicator rather than as a complete decomposition of every possible uncertainty source. Recent studies have continued to refine evidential uncertainty modeling, particularly for out-of-distribution detection [28].

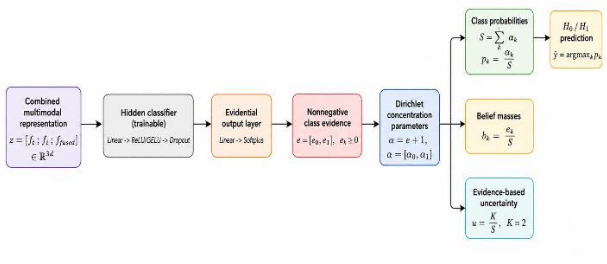


Fig. 3. End-to-end Dirichlet evidential decision head. The combined multimodal representation is processed by a trainable hidden classifier and evidential output layer to produce nonnegative class evidence. The evidence is converted into Dirichlet concentration parameters and subsequently decomposed into class probabilities, belief masses, and an evidence-based uncertainty estimate.

F. End-to-End Training Objective and Decision Rule

The complete framework is trained end to end using an objective that combines evidential classification, class-aware focal reweighting, unsupported-evidence regularization, and pairwise ranking.

For a one-hot label vector $\mathbf{y}$, the basic evidential loss consists of the expected squared error and the Dirichlet predictive variance:

$$\mathcal{L}_{\mathrm{EDL}} = \sum_{k=1}^{K} \left[(y_k - p_k)^2 + \frac{p_k(1-p_k)}{S+1} \right]. \tag{27}$$

The first term penalizes the discrepancy between the target and expected class probability, whereas the second term accounts for the predictive variance induced by the Dirichlet distribution. An incorrect prediction with large evidence therefore receives a stronger penalty than an uncertain prediction.

To emphasize difficult observations, particularly weak H_1 samples, the evidential loss is reweighted using class-aware focal modulation:

$$\mathcal{L}_{\mathrm{focal}} = w_y(1 - p_y)^\gamma \mathcal{L}_{\mathrm{EDL}}, \tag{28}$$

where $w_y$is the weight of the ground-truth class and $\gamma$is the focusing parameter.

Evidence assigned to non-target classes is regularized toward a uniform Dirichlet prior. The modified concentration vector is defined as

$$\tilde{\boldsymbol{\alpha}} = \mathbf{y} + (1 - \mathbf{y}) \odot \boldsymbol{\alpha}, \tag{29}$$

and the corresponding regularization term is

$$\mathcal{L}_{\mathrm{KL}} = D_{\mathrm{KL}}[\mathrm{Dir}(\tilde{\boldsymbol{\alpha}}) \parallel \mathrm{Dir}(\mathbf{1})]. \tag{30}$$

The operation in (29) preserves the concentration of the ground-truth class while regularizing unsupported evidence assigned to the non-target class.

Because spectrum-sensing performance depends strongly on the score separation between occupied and idle observations, the sensing score is defined as

$$s(\mathbf{x}) = \log p_1(\mathbf{x}) - \log p_0(\mathbf{x}). \tag{31}$$

For the positive and negative sample sets $\mathcal{P}$and $\mathcal{N}$, respectively, the pairwise ranking loss is

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}||\mathcal{N}|}\sum_{i\in\mathcal{P}}\sum_{j\in\mathcal{N}}\max[0, m-(s_i-s_j)], \tag{32}$$

where $m$is the ranking margin. This term encourages occupied observations to receive higher sensing scores than idle observations.

The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}} + \lambda_{\text{AUC}}\mathcal{L}_{\text{rank}}. \tag{33}$$

The gradients of $\mathcal{L}$are propagated through the evidential output layer, hidden classifier, cross-modal refinement module, temporal encoder, and image encoder. Therefore, all trainable components, including $\mathbf{W}_e$and $\mathbf{b}_e$in (21), are jointly updated during every optimization step.

During inference, the final sensing decision is determined by

$$\hat{y} = \begin{cases} 1, & p(H_1 \mid \mathbf{x}) \geq \tau, \\ 0, & p(H_1 \mid \mathbf{x}) < \tau, \end{cases} \tag{34}$$

where $\tau$is selected exclusively using the validation set and subsequently fixed for all test sets. The threshold-selection objective is

$$\begin{aligned} J(\tau) = \quad & 0.20\,\text{ACC} + 0.28\,\text{F1} + 0.12\,\text{Precision} \\ & +0.28\,P_d + 0.12\,\text{AUC} - 0.06\,P_{\text{fa}} \\ & -1.2\max(0, P_{\text{fa}} - 0.035). \end{aligned} \tag{35}$$

The objective in (35) emphasizes the detection of occupied observations while imposing a soft penalty on excessive false alarms. No test-set labels are used during threshold selection.

IV. Performance evaluation

This section evaluates the proposed framework from three perspectives: in-distribution sensing performance, generalization to unseen SNR conditions, and blind sensing of unseen modulation formats. The low-SNR transition behavior is further examined to determine whether improved detection is achieved without excessive false alarms.

A. Experimental Setup

(1) Spectrum-Sensing Dataset Construction

RadioML 2018.01A was originally developed for automatic modulation classification rather than binary spectrum sensing [19]. It contains 2,555,904 complex baseband sequences from 24 modulation formats. Each observation consists of 1,024 IQ samples and is associated with an SNR value ranging from $-20$to 30dB in 2-dB increments.

To construct a spectrum-sensing benchmark, the original observations are reorganized into positive and negative sensing classes. In the conventional physical formulation, H_1 denotes the presence of a primary-user signal, whereas H_0 denotes an idle channel containing noise only. In the implemented benchmark, positive samples are selected from modulated RadioML observations, while the negative class contains synthetically generated complex AWGN and RadioML observations at -20dB. The latter are operationally treated as noise-dominated hard-negative surrogates because their signal components are almost completely submerged by noise. This assignment is an experimental labeling convention rather than a strict physical realization of H_0.

Stratified sampling is performed jointly over modulation format and SNR. For each modulation–SNR combination, 76 observations are selected, yielding 1,976 observations per modulation and 47,424 selected RadioML observations in total. Fourteen modulation formats are used as known formats, whereas the remaining ten are reserved exclusively for blind-modulation evaluation.

The evaluation consists of three stages: in-distribution validation, unseen-SNR testing, and unseen-modulation testing. The training partition is listed separately in Table 1 for completeness.

Table 1. Spectrum-sensing dataset construction and evaluation protocol

Partition	Positive samples	Negative samples	Primary objective
Training	9,363 known-format samples, $[-10, 10]$ dB	851 hard negatives and 8,512 AWGN samples	Model optimization
Validation	2,341 known-format samples, $[-10, 10]$ dB	213 hard negatives and 2,128 AWGN samples	Model selection and threshold calibration
Test Set 1	14,896 known-format samples, $[-18, -12] \cup [12, 30]$ dB	14,896 AWGN samples	Unseen-SNR generalization
Test Set 2	19,000 unseen-format samples, $[-18, 30]$ dB	760 hard negatives and 18,240 AWGN samples	Blind-modulation generalization

Known formats: BPSK, QPSK, 8PSK, 16PSK, OOK, 4ASK, 8ASK, 16QAM, 32QAM, 64QAM, AM-DSB-SC, AM-SSB-SC, GMSK, and OQPSK.

Unseen formats: 32PSK, 16APSK, 32APSK, 64APSK, 128APSK, 128QAM, 256QAM, AM-SSB-WC, AM-DSB-WC, and FM. All SNR intervals use a step size of 2dB.

Test Set 1 retains the modulation families used during training but evaluates only SNR values outside the training interval. It therefore isolates SNR-domain generalization. Test Set 2 contains positive samples exclusively from modulation formats excluded from training and evaluates whether the detector learns general signal-presence characteristics rather than modulation-specific templates.

(2) Baseline Methods

All baseline models are trained and evaluated using the same data partitions and evaluation protocol as the proposed method.

- **CNN-based method**: Following the general CNN-based spectrum-sensing paradigm in [5], [6], an image-only baseline is implemented using the same eight signal-image modalities, including PSD, STFT, GASF, GADF, MTF, MDF, RP, and RPM. The eight $\mathbf{32 \times 32}$channels are processed by an ImageNet-pretrained ResNet18 whose first convolutional layer is modified to accept eight-channel inputs. The extracted feature is mapped to the two sensing classes through a dropout-assisted linear classifier. This baseline isolates the contribution of image-domain spectral, transition, and recurrence structures without temporal modeling or cross-modal interaction.

- **BiLSTM-based method**: Motivated by recurrent spectrum-sensing architectures such as [7], a temporal-only BiLSTM baseline is constructed from four standardized sequential features, namely the in-phase component, quadrature component, instantaneous amplitude, and unwrapped phase. A two-layer bidirectional LSTM with 64 hidden units in each direction models forward and backward dependencies over the 1,024-sample sequence. The output at the final time step is passed through a two-layer fully connected classifier with ReLU activation and dropout. This baseline evaluates the effectiveness of temporal waveform evolution without image-domain information.

- **BCAM-Net**: BCAM-Net is reproduced following the multimodal design in [8]. Its temporal branch uses normalized magnitude and phase sequences and employs a two-layer bidirectional GRU to extract sequential features. The time-frequency branch transforms each IQ observation into two-channel complex CWT maps and processes them using residual convolutional blocks equipped with channel and spatial attention. The resulting temporal and time-frequency embeddings are refined through bidirectional multihead cross-modal interaction, after which the original and enhanced representations are concatenated for binary sensing. BCAM-Net therefore serves as the principal multimodal baseline for evaluating the effectiveness of the proposed gated cross-modal refinement strategy.

(3) Implementation Details

Each IQ observation is converted into a six-channel temporal representation,

$$[I, Q, A, \phi, \Delta\phi, f_{\text{inst}}], \tag{36}$$

where A, ϕ, $\Delta\phi$, and f_{inst} denote instantaneous amplitude, unwrapped phase, phase difference, and normalized instantaneous frequency, respectively.

The image modality contains PSD, STFT, GASF, GADF, MTF, MDF, RP, and RPM channels. Each channel is independently normalized and resized to 128×128.

Training-only augmentation includes phase rotation, circular time shifting, carrier-frequency offset, IQ imbalance, amplitude scaling, and additive-noise perturbation. Additional low-SNR positive views are generated to improve weak-signal robustness. Validation and test observations are not augmented.

The proposed model is trained using AdamW with a batch size of 64. The maximum number of epochs is 55, including five warm-up epochs followed by cosine learning-rate decay. Exponential moving averaging with a decay factor of 0.995, gradient clipping, and early stopping with a patience of 12 are employed. Early stopping selects the EMA checkpoint at epoch 37. The complete model contains approximately 11.89 million trainable parameters.

The training objective combines evidential classification, focal reweighting, KL regularization, and pairwise AUC ranking. For the proposed model, the validation-calibrated threshold is 0.560. Each baseline is likewise calibrated exclusively using its own validation predictions, after which the corresponding threshold is fixed for both test sets. No test labels are used for operating-point selection.

All experiments are implemented in PyTorch and conducted on an NVIDIA Tesla T4 GPU.

(4) Evaluation Metrics

The reported metrics include accuracy, F1-score, precision, detection probability P_d, false-alarm probability P_{fa}, and AUC. The sensing-specific metrics are

$$P_d = \frac{\text{TP}}{\text{TP} + \text{FN}}, P_{\text{fa}} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{37}$$

AUC evaluates threshold-independent ranking quality, whereas P_d and P_{fa} characterize the detector at the selected operating point. In practical spectrum sensing, a high P_d is useful only when P_{fa} remains sufficiently low.

For the proposed evidential model, the uncertainty mass is calculated as

$$u = \frac{K}{\sum_{k=1}^{K} \alpha_k}, \tag{38}$$

where $K = 2$ is the number of sensing classes. Mean uncertainty and the uncertainties of correct and incorrect predictions are additionally reported to assess whether the evidential output can identify unreliable decisions.

B. Comparison Experiments

Table 2. Performance comparison of different models on three evaluation sets.

Model	Metric	Validation	Test Set 1	Test Set 2
Ours	ACC	**0.9507**	**0.8530**	**0.8890**
	F1-score	**0.9490**	**0.8316**	**0.8779**
	Precision	0.9813	0.9732	0.9752
	Recall (Pd)	**0.9188**	0.7260	0.7983
	Pfa	0.0175	0.0200	0.0203
	AUC	**0.9805**	**0.8752**	**0.9166**
BCAM-Net	ACC	0.8642	0.8206	0.8362
	F1-score	0.8481	0.7896	0.8114
	Precision	0.9616	0.9545	0.9559
	Recall (Pd)	0.7587	0.6733	0.7049
	Pfa	0.0303	0.0321	0.0325
	AUC	0.9256	0.8590	0.8922
CNN	ACC	0.8379	0.7862	0.8075
	F1-score	0.8446	0.7850	0.8106
	Precision	0.8112	0.7895	0.7978
	Recall (Pd)	0.8808	**0.7805**	**0.8238**
	Pfa	0.2050	0.2081	0.2088
	AUC	0.9304	0.8615	0.8923
BiLSTM	ACC	0.9231	0.8526	0.8808
	F1-score	0.9180	0.8304	0.8674
	Precision	**0.9834**	**0.9778**	**0.9776**
	Recall (Pd)	0.8607	0.7216	0.7795
	Pfa	**0.0145**	**0.0164**	**0.0178**
	AUC	0.9644	0.8650	0.9074

(1) Overall Quantitative Comparison

e proposed model achieves the best ACC, F1-score, and AUC on all three evaluation sets. On the validation set, it obtains an ACC of 0.9507, an F1-score of 0.9490, and an AUC of 0.9805, while maintaining $P_d = 0.9188$and $P_{\text{fa}} = 0.0175$.

Compared with BCAM-Net, the proposed model improves validation ACC, F1-score, and AUC by 8.65, 10.09, and 5.49 percentage points, respectively. Since BCAM-Net also uses multimodal inputs, the improvement cannot be explained merely by the number of signal representations. Instead, it is consistent with the combined effects of heterogeneous temporal and image features, controlled bidirectional refinement, and evidential training.

CNN obtains a relatively high validation $P_d$of 0.8808, but its P_{fa} reaches 0.2050. This indicates that image-domain structures alone may respond strongly to signal-like patterns generated by noise. BiLSTM provides the strongest single-modal baseline, but its validation $P_d$remains lower than that of the proposed model. The proposed framework therefore achieves a more favorable balance between occupied-signal detection and false-alarm control.

(2) Generalization Under Unseen SNRs

On Test Set 1, the proposed model achieves an ACC of 0.8530, an F1-score of 0.8316, and an AUC of 0.8752. It retains a precision of 0.9732, with $P_d = 0.7260$ and $P_{\text{fa}} = 0.0200$.

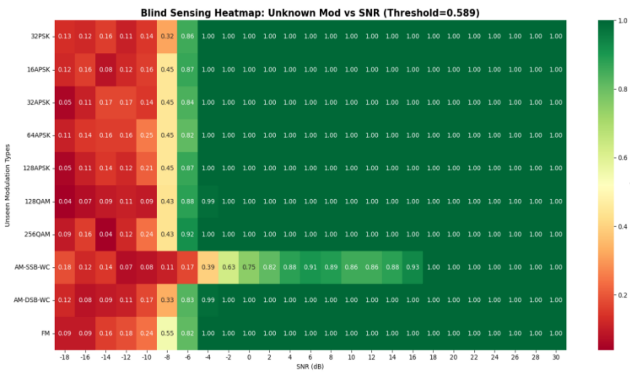
Compared with BCAM-Net, the proposed method improves ACC, F1-score, and AUC by 3.24, 4.20, and 4.72 percentage points, respectively. Compared with BiLSTM, the improvement at the selected operating point is modest, but the AUC increases by 1.02 percentage points. Since AUC evaluates score ordering across all possible thresholds, this result indicates stronger positive–negative separability under SNR-domain shift.

CNN achieves a higher $P_d$of 0.7805 but also produces $P_{\text{fa}} = 0.2081$. Its detection gain therefore comes at the cost of frequent false occupancy declarations. In contrast, the proposed method maintains a false-alarm probability close to 2%.

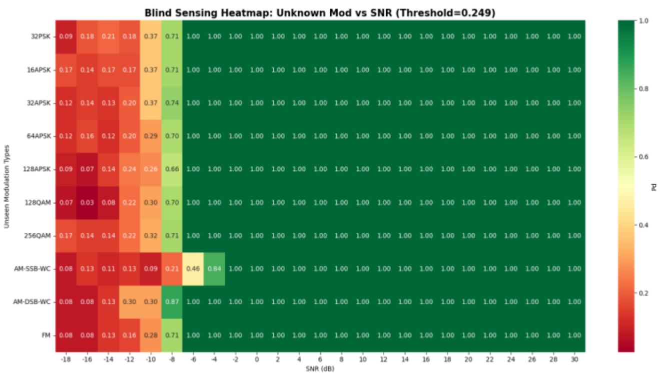
The observed robustness is consistent with the combined use of low-SNR augmentation, explicit phase-frequency features, and pairwise ranking supervision. These factors may improve the representation of weak occupied observations and preserve score separation outside the training SNR interval.

(3) Low-SNR Transition and Blind-Modulation Generalization

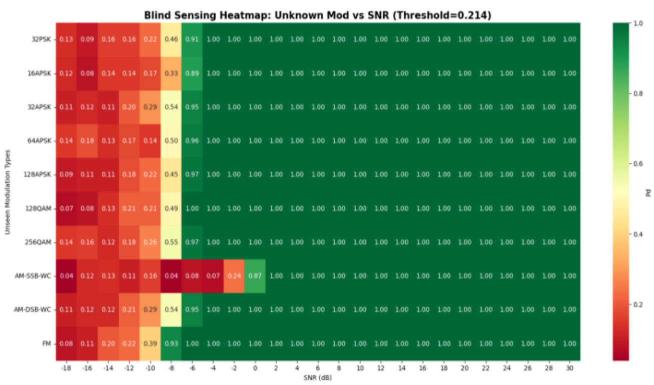
Test Set 2 is the most demanding evaluation stage because all positive observations belong to modulation formats excluded from training. It therefore evaluates whether the detector learns transferable signal-presence characteristics rather than relying on closed-set modulation templates.



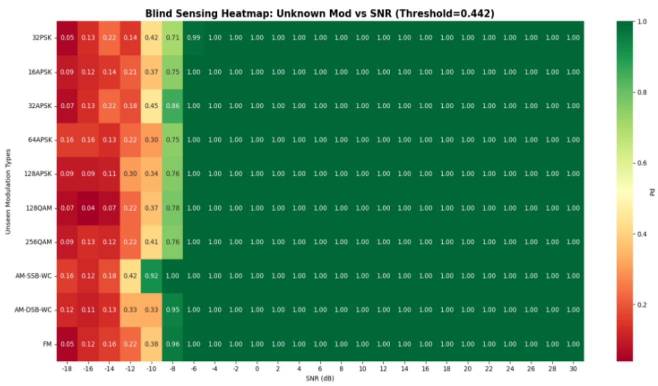
(a) CNN



(b) BiLSTM



(c) BCAM-Net



(d) Ours

Fig. 4. Modulation-wise detection probability $P_d$heatmaps on Test Set 2 for unseen modulation formats: (a) CNN, (b) BiLSTM, (c) BCAM-Net, and (d) the proposed method. All panels use identical modulation and SNR orders, a shared color scale, and thresholds calibrated on the corresponding validation sets.

The proposed model achieves an ACC of 0.8890, an F1-score of 0.8779, $P_d = 0.7983$, $P_{\text{fa}} = 0.0203$, and an AUC of 0.9166. Compared with BCAM-Net, it improves ACC, F1-score, P_d, and AUC by 5.28, 6.65, 9.34, and 2.44 percentage points, respectively. Compared with BiLSTM, it improves ACC by 0.82 percentage points and AUC by 0.92 percentage points.

CNN obtains a slightly higher P_d of 0.8238, but its P_{fa} reaches 0.2088. Its higher recall is therefore accompanied by substantially more false occupancy declarations. The proposed model provides a more balanced operating point by preserving competitive detection performance while reducing the false-alarm probability by 18.85 percentage points.

e heatmaps reveal different low-SNR transition patterns. BCAM-Net exhibits an uneven transition across modulation families, whereas BiLSTM performs well on several digital modulations but is less consistent for waveform families with unfamiliar temporal structures. CNN becomes strongly activated at moderate SNRs, but its high global P_{fa} indicates substantial over-detection.

The proposed method is expected to exhibit a more coherent transition across unseen modulation formats while maintaining a false-alarm probability close to 2%. This behavior may be attributed to the complementary information supplied by temporal phase-frequency dynamics and image-domain spectral, transition, and recurrence structures. The learnable gates further restrict the magnitude of cross-modal residual updates and reduce the risk that a severely corrupted representation dominates the fused feature.

The heatmaps must be interpreted jointly with P_{fa}. An aggressive detector may reach a high P_d at very low SNRs while producing an unacceptable number of false alarms. Consequently, early heatmap activation alone does not imply a superior sensing model.

Overall, the proposed framework improves in-distribution performance, preserves stronger score separability under SNR shift, and generalizes more reliably to unseen modulation formats. Its main advantage is not simply a higher recall, but a more stable balance among P_d, P_{fa}, and AUC under different distribution shifts.

The present evaluation remains limited to simulated RadioML observations, and the -20dB samples are operationally treated as noise-dominated hard negatives. Future experiments will incorporate independently recorded idle-channel noise, over-the-air measurements, and hardware-impaired signals. Ablation and calibration analyses will also be conducted to isolate the contribution of individual components and further evaluate the evidential uncertainty output.

V. Conclusion

This paper presented an evidential multimodal framework for reliable binary spectrum sensing under SNR and modulation distribution shifts. Complementary temporal and signal-image representations are jointly modeled through dedicated feature encoders and a gated bidirectional cross-modal refinement module, while Dirichlet evidential learning provides spectrum-occupancy predictions together with an evidence-based uncertainty estimate.

Experiments under in-distribution validation, unseen-SNR testing, and unseen-modulation testing demonstrate that the proposed framework consistently achieves the best ACC, F1-score, and AUC among the compared methods while maintaining a low false-alarm probability. The results further show that reliable spectrum sensing should not be assessed solely by in-distribution classification accuracy. Robust sensing requires stable score separability, controlled false alarms, and transferable signal-presence detection under distribution shifts. From this perspective, the proposed framework provides a promising approach to trustworthy physical-layer sensing by combining complementary signal representations with uncertainty-aware decision modeling.

The current evaluation is limited to simulated RadioML observations, and the -20-dB samples are operationally treated as noise-dominated hard negatives. Future work will therefore investigate over-the-air measurements, independently recorded idle-channel noise, hardware impairments, and more complex interference environments. Further studies will also consider uncertainty calibration, online adaptation, and lightweight implementations for practical cognitive-radio deployment.

References

[1] J. Mitola and G. Q. Maguire, “Cognitive radio: making software radios more personal,” IEEE Personal Communications, vol. 6, no. 4, pp. 13–18, 1999.

[2] S. Haykin, “Cognitive radio: brain-empowered wireless communications,” IEEE Journal on Selected Areas in Communications, vol. 23, no. 2, pp. 201–220, 2005.

[3] T. Yücek and H. Arslan, “A survey of spectrum sensing algorithms for cognitive radio applications,” IEEE Communications Surveys & Tutorials, vol. 11, no. 1, pp. 116–130, 2009.

[4] J. Lundén, V. Koivunen, A. Huttunen, and H. V. Poor, “Spectrum sensing in cognitive radios based on multiple cyclic frequencies,” IEEE Transactions on Signal Processing, vol. 57, no. 12, pp. 4654–4666, 2009.

[5] W. Lee, M. Kim, and D.-H. Cho, “Deep cooperative sensing: cooperative spectrum sensing based on convolutional neural networks,” IEEE Transactions on Vehicular Technology, vol. 68, no. 3, pp. 3005–3015, 2019.

[6] K. Tekbiyik, O. Akbunar, A. R. Ekti, A. Gorcin, G. K. Kurt, and K. A. Qaraqe, “Spectrum sensing and signal identification with deep learning based on spectral correlation function,” IEEE Transactions on Vehicular Technology, vol. 70, no. 10, pp. 10514–10527, 2021.

[7] A. Kumar, A. Nanthaamornphong, and M. Masud, “RNN-Bi-LSTM spectrum sensing algorithm for NOMA waveform with diverse channel conditions,” Scientific Reports, vol. 15, 31022, 2025.

[8] Y. Han, Z. Li, A. Gendia, T. Ide, and O. Muta, “BCAM-Net: A bidirectional cross-attention multimodal network for IoT spectrum sensing under generalized Gaussian noise,” Computers, Materials & Continua, vol. 87, no. 2, 2026.

[9] M. Sensoy, L. Kaplan, and M. Kandemir, “Evidential deep learning to quantify classification uncertainty,” in Advances in Neural Information Processing Systems, 2018.

[10] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in Proceedings of the 34th International Conferece on Machine Learning, vol. 70, pp. 1321–1330, 2017.

[11] Y. Ovadia et al., “Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift,” in Advances in Neural Information Processing Systems, vol. 32, 2019.

[12] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian approximation: Representing model uncertainty in deep learning,” in Proceedings of the 33rd International Conference on Machine Learning, pp. 1050–1059, 2016.

[13] Z. Chen, Y.-Q. Xu, H. Wang, and D. Guo, “Deep STFT-CNN for spectrum sensing in cognitive radio,” IEEE Communications Letters, vol.

25, no. 3, pp. 864–868, Mar. 2021, doi: 10.1109/LCOMM.2020.3037273.

[14] P. Zhen, B. Zhang, Z. Chen, D. Guo, and W. Ma, “Spectrum sensing method based on wavelet transform and residual network,” IEEE Wireless Communications Letters, vol. 11, no. 12, pp. 2517–2521, Dec. 2022, doi: 10.1109/LWC.2022.3207296.

[15] X. Fang, M. Jin, Q. Guo, and T. Jiang, “CNN-Transformer-based cooperative spectrum sensing in cognitive radio networks,” IEEE Wireless Communications Letters, vol. 14, no. 5, pp. 1576–1580, May 2025, doi: 10.1109/LWC.2025.3550536.

[16] Z. Gao, Y. Li, Z. Chen, M. Asif, L. Xu, X. Li, et al., “Intelligent spectrum sensing of consumer IoT based on GAN-GRU-YOLO,” IEEE Transactions on Consumer Electronics, vol. 70, no. 3, pp. 6140–6148, Aug. 2024, doi: 10.1109/TCE.2024.3418103.

[17] W. Zhang, Y. Wang, X. Chen, Z. Cai, and Z. Tian, “Spectrum Transformer: An attention-based wideband spectrum detector,” IEEE Transactions on Wireless Communications, vol. 23, no. 9, pp. 12343–12353, Sep. 2024, doi: 10.1109/TWC.2024.3391515.

[18] H. Xi, W. Guo, Y. Yang, R. Yuan, and H. Ma, “Cross-attention mechanism-based spectrum sensing in generalized Gaussian noise,” Scientific Reports, vol. 14, Art. no. 23261, 2024, doi: 10.1038/s41598-024-74341-4.

[19] Z. Wang and T. Oates, “Imaging time-series to improve classification and imputation,” in Proc. 24th Int. Joint Conf. Artificial Intelligence (IJCAI), 2015, pp. 3939–3945.

[20] J.-P. Eckmann, S. O. Kamphorst, and D. Ruelle, “Recurrence plots of dynamical systems,” Europhys. Lett., vol. 4, no. 9, pp. 973–977, 1987.

[21] C. Lea, M. D. Flynn, R. Vidal, A. Reiter, and G. D. Hager, “Temporal convolutional networks for action segmentation and detection,” in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2017, pp. 156–165.

[22] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778.

[23] A. Nagrani, S. Yang, A. Arnab, A. Jansen, C. Schmid, and C. Sun, “Attention bottlenecks for multimodal fusion,” in Advances in Neural Information Processing Systems, vol. 34, 2021.

[24] Y. Wang, X. Chen, L. Cao, W. Huang, F. Sun, and Y. Wang, “Multimodal token fusion for vision transformers,” in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2022, pp. 12186–12195.

[25] Z. Han, C. Zhang, H. Fu, and J. T. Zhou, “Trusted multi-view classification with dynamic evidential fusion,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 45, no. 2, pp. 2551–2566, Feb. 2023, doi: 10.1109/TPAMI.2022.3171983.

[26] W. Liu, Y. Chen, X. Yue, C. Zhang, and S. Xie, “Safe multi-view deep classification,” in Proc. AAAI Conf. Artificial Intelligence, vol. 37, no. 7, pp. 8870–8878, 2023, doi: 10.1609/aaai.v37i7.26066.

[27] C. Xu, J. Si, Z. Guan, W. Zhao, Y. Wu, and X. Gao, “Reliable conflictive multi-view learning,” in Proc. AAAI Conf. Artificial Intelligence, vol. 38, no. 14, pp. 16129–16137, 2024, doi: 10.1609/aaai.v38i14.29546.

[28] J. Qu, Y. Chen, X. Yue, W. Fu, and Q. Huang, “Hyper-opinion evidential deep learning for out-of-distribution detection,” in Advances in Neural Information Processing Systems, vol. 37, 2024.